Neo N3 RISC-V Virtual Machine

Formal Specification and Architecture

Version 1.0

Neo RISC-V VM Project

`github.com/jim8y/neo-riscv-vm`

`github.com/neo-project/neo` — *feature/riscv-native-contracts*

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Abstract

This document specifies the RISC-V Virtual Machine solution for Neo N3, based on PolkaVM v0.32.0 (Parity Technologies). The solution introduces RISC-V as the primary execution engine while maintaining complete backward compatibility with all existing NeoVM smart contracts through an interpreter bridge.

We describe the dual-contract architecture in which both NeoVM bytecode and native RISC-V binaries coexist on a unified execution infrastructure. The specification covers: the four-layer system architecture, the contract type system with automatic binary detection, the deployment and invocation models, the complete SYSCALL interop surface (40 registered handlers with hash definitions), the gas metering system, the storage model, the security sandbox, the transaction model, and the NEF file format.

All 983 existing Neo N3 integration tests—including 175 native contract tests and 40 SYSCALL interop tests—pass without modification, demonstrating full compatibility with the existing transaction model, native contracts, storage system, and interop infrastructure. The changes to Neo core are limited to a single field addition (`ContractState.Type`) and a dispatch branch (`CallContract`), totaling approximately 50 lines of production code.

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1 Introduction

1.1 Background and Motivation

Neo N3 executes smart contracts using the NeoVM, a stack-based virtual machine with approximately 130 opcodes. While NeoVM provides a deterministic and secure execution environment, its interpreted nature imposes performance limitations on compute-intensive operations including cryptographic primitives, complex data processing, and algorithmic workloads.

The blockchain industry has increasingly adopted WebAssembly (WASM) and RISC-V as execution targets for smart contract virtual machines, offering near-native performance while maintaining sandboxed execution guarantees. This specification describes the integration of PolkaVM, a production-grade sandboxed RISC-V virtual machine developed by Parity Technologies for use in Polkadot, as the execution backbone for Neo N3.

1.2 Dual-Execution Architecture

The solution provides two execution paths on a unified infrastructure:

1. **NeoVM Path**: Existing NeoVM bytecode is interpreted by a NeoVM interpreter running as a guest program inside the PolkaVM sandbox. This path preserves complete backward compatibility with all existing contracts, tools, and SDKs.
2. **RISC-V Path**: Native RISC-V binaries are executed directly by PolkaVM, providing near-native performance for new contracts compiled from Rust, C, C++, or other languages targeting the RISC-V ISA.

Both paths share identical deployment infrastructure, invocation models, SYSCALL interop surfaces, gas metering, storage namespaces, and notification systems.

1.3 Design Principles

1. **Backward Compatibility**: All existing NeoVM contracts, tools, SDKs, wallets, and infrastructure continue to work without modification. Zero protocol changes are required.
2. **Transparent Dispatch**: The execution backend is determined automatically at deployment time. Callers invoke both contract types identically via `System.Contract.Call`.
3. **Unified Interop**: Both execution paths use the same 40 SYSCALL handlers, providing identical access to runtime, storage, contract, and cryptographic operations.
4. **Sandbox Security**: PolkaVM provides memory isolation, deterministic execution, and syscall-gated host interaction with no direct host memory access.
5. **Minimal Surface Area**: Changes to Neo core are limited to a single field addition and a dispatch branch, minimizing the attack surface and review burden.

1.4 Document Scope

This specification covers:

- System architecture and layered model (Section 2)
- Contract type system and binary detection (Section 3)
- Deployment and update model (Section 4)
- Contract invocation and cross-contract calls (Section 5)
- Complete SYSCALL interface with 40 handlers (Section 6)
- Gas metering and pricing (Section 7)
- Storage model and iterators (Section 8)
- Security model and sandbox guarantees (Section 9)
- Transaction model and witness verification (Section 10)

- NEF file format (Section 11)
- Native contract compatibility (Section 12)
- Notification and event system (Section 13)
- Implementation details (Section 14)
- Backward compatibility analysis (Section 15)
- Migration path (Section 16)

2 System Architecture

2.1 Layered Model

The system consists of four layers, each with well-defined responsibilities and interfaces, as shown in Figure 1.

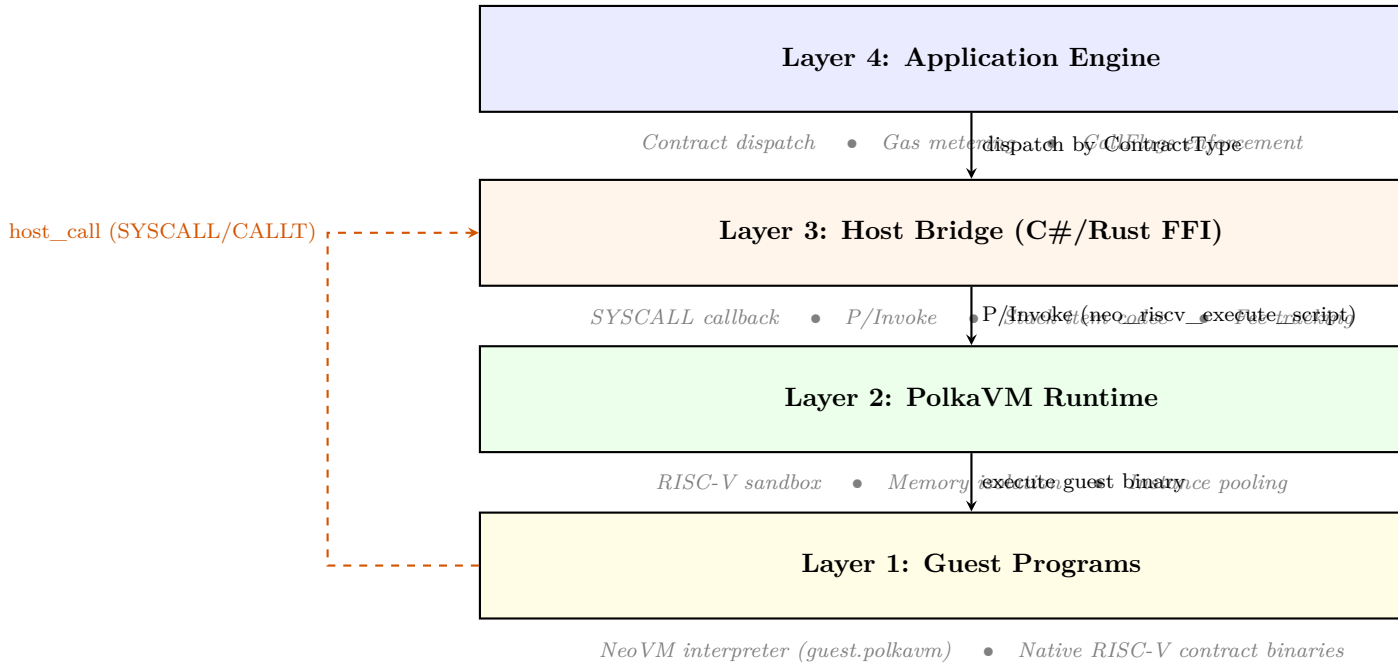

Figure 1: Four-layer execution architecture with data flow

2.1.1 Layer 1: Guest Programs

Two types of guest programs execute inside the PolkaVM sandbox:

- **NeoVM Interpreter** (`guest.polkavm`): A `no_std` Rust program implementing the complete NeoVM instruction set (130+ opcodes). It receives NeoVM bytecode via PolkaVM's auxiliary data mechanism, interprets it, and communicates with the host through `host_call` SYSCALLs. The interpreter manages its own evaluation stack, local variables, static fields, call stack, alt stack, and exception handling frames internally.
- **Native RISC-V Binaries**: Contract-specific PolkaVM binaries compiled from high-level languages (Rust, C, C++). They interact with the host through the identical `host_call` SYSCALL interface, gaining access to the same 40 SYSCALL handlers.

2.1.2 Layer 2: PolkaVM Runtime

PolkaVM provides a deterministic, sandboxed execution environment with the following properties:

- **Fixed Memory Arena**: Each guest operates within a configurable memory arena (256 MB default, 4 MB used for interpreter). No dynamic expansion beyond the configured limit.
- **No Direct Host Access**: Guest programs cannot read or write host memory. All data exchange occurs through the `host_call` mechanism with explicit buffer pointers.
- **Deterministic**: No floating-point arithmetic, no non-deterministic system calls, no timing-dependent operations.
- **Instance Pooling**: PolkaVM instances are pooled and reused across executions, amortizing the cost of sandbox initialization.

2.1.3 Layer 3: Host Bridge

The C#–Rust bridge manages all inter-layer communication:

- **P/Invoke**: C# calls Rust host library functions (`neo_riscv_execute_script_with_host`).
- **SYSCALL Callback**: Guest `host_call` invocations are routed to C# `HandleHostCallback`, which dispatches by SYSCALL hash to the appropriate handler.
- **Stack Item Codec**: Bidirectional serialization between guest `StackValue` (postcard) and C# `StackItem` (native struct).
- **Fee Tracking**: Gas consumed by the guest is accumulated and reported back to the ApplicationEngine.

2.1.4 Layer 4: Application Engine

The Neo N3 ApplicationEngine (extended as `RiscvApplicationEngine`) handles contract dispatch:

- Contract type detection (NeoVM vs. RISC-V via `ContractState.Type`)
- SYSCALL interop dispatch (40 registered handlers)
- Native contract method resolution via reflection
- Gas metering and enforcement
- Storage access control and caching
- Notification emission

2.2 Execution Flow

Figure 2 shows the contract execution flow, from transaction submission through to result delivery.

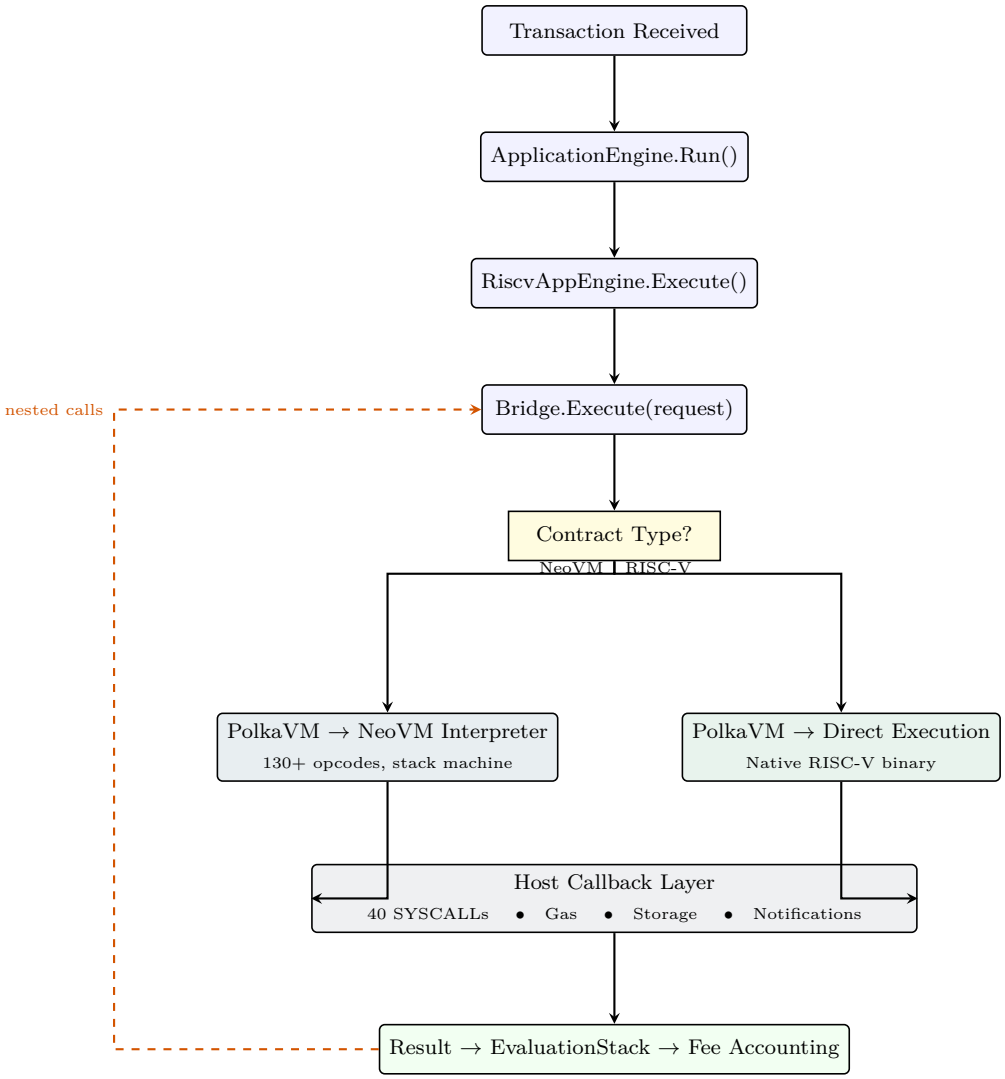

Figure 2: Contract execution flow with nested call support

2.3 Wire Protocol

Data exchange between the guest and host uses two serialization formats:

1. **Postcard Codec** (guest ↔ host): Stack items are serialized using the `postcard` binary format. Tags: 0=Integer, 1=BigInteger, 2=ByteString, 3=Boolean, 4=Array, 5=Struct, 6=Map, 7=Interop, 8=Iterator, 9=Null.
2. **Native Struct** (Rust ↔ C#): The FFI layer uses `NativeStackItem` structs with kind discriminants (0=Integer, 1=ByteString, 2=Null, 3=Boolean, 4=Array, 5=BigInteger, 6=Iterator, 7=Struct, 8=Map, 9=Interop).

3 Contract Type System

3.1 ContractType Enumeration

Listing 1: ContractType enumeration

```
/// <summary>
/// Defines the execution backend for a deployed contract.
/// </summary>
```

```
public enum ContractType : byte
{
    /// <summary>
    /// NeoVM bytecode executed by the NeoVM interpreter guest inside
        PolkaVM.
    /// This is the default. All existing contracts have this type.
    /// </summary>
    NeoVM = 0,

    /// <summary>
    /// PolkaVM RISC-V binary executed directly by PolkaVM.
    /// The NEF script begins with the PVM\0 magic bytes.
    /// </summary>
    RiscV = 1,
}
```

3.2 ContractState Extension

The `ContractState` structure is extended with a single field. The field is positioned after `UpdateCounter` for serialization compatibility:

Listing 2: ContractState with Type field

```
public class ContractState : IInteroperableVerifiable
{
    public int Id;
    public ushort UpdateCounter;
    public ContractType Type;       // NEW: execution backend type
    public required UInt160 Hash;
    public required NefFile Nef;
    public required ContractManifest Manifest;
    public ReadOnlyMemory<byte> Script => Nef.Script;
}
```

3.3 Binary Format Detection

A contract's execution backend is determined by inspecting the first 4 bytes of the NEF script for the PolkaVM magic signature:

Offset	Value	ASCII	Description
0	`0x50`	`P`	PolkaVM identifier, byte 1
1	`0x56`	`V`	PolkaVM identifier, byte 2
2	`0x4D`	`M`	PolkaVM identifier, byte 3
3	`0x00`	`\0`	Null terminator

Table 1: PVM\0 magic bytes for RISC-V binary detection

Detection is unambiguous: NeoVM bytecode never begins with these four bytes, as the first byte (`0x50`) corresponds to no valid NeoVM opcode in the push range. Detection is performed during deployment and update.

3.4 Serialization

3.4.1 StackItem Representation

The `Type` field is serialized as the 6th element of the `ContractState` StackItem array representation:

$$\texttt{Array}\Big[\ Id,\ \ UpdateCounter,\ \ Hash,\ \ Nef,\ \ Manifest,\ \ \underbrace{(\texttt{int})(\texttt{byte})Type}_{\text{NEW}}\ \Big]$$

3.4.2 Backward-Compatible Deserialization

During deserialization, if the array contains fewer than 6 elements (i.e., serialized before the `Type` field was added):

$$Type = \begin{cases} \text{ContractType.NeoVM} & \text{if } \texttt{Array.Count} \leq 5 \\ (\texttt{ContractType})(\texttt{byte})\texttt{Array}[5] & \text{otherwise} \end{cases}$$

This ensures all existing serialized `ContractState` data remains valid without requiring a storage migration.

3.4.3 JSON Representation

The JSON output includes the type as a string field:

Listing 3: ContractState JSON representation

```
{
    "id": -1,
    "updatecounter": 0,
    "type": "NeoVM",
    "hash": "0xfffdc93764dbaddd97c48f252a53ea4643faa3fd",
    "nef": {
        "magic": 860243278,
        "compiler": "neo-core-v3.0",
        "source": "",
        "tokens": [],
        "script": "EEEa93tn...",
        "checksum": 3581846399
    },
    "manifest": { ... }
}
```

3.5 Deterministic Hash

The contract hash computation is independent of the `Type` field:

$$\text{hash} = \text{RIPEMD-160}(\text{SHA-256}(\texttt{0x00} \parallel \text{sender} \parallel \text{nefChecksum}_{\text{LE}} \parallel \text{name}_{\text{UTF8}})) \tag{1}$$

This ensures hash stability across type changes during contract updates and across all Neo implementations.

4 Deployment Model

4.1 Deploy Algorithm

Algorithm 1 ContractManagement.Deploy

Require: nefFile $\neq \emptyset$, manifest $\neq \emptyset$
Require: tx $\neq$ null, fee $\geq$ minDeploymentFee

```
nef ← NefFile.Deserialize(nefFile)
parsedManifest ← ContractManifest.Parse(manifest)
ValidateScript(nef.Script, parsedManifest.Abi)
hash ← GetContractHash(tx.Sender, nef.Checksum, parsedManifest.Name)
assert ¬IsBlocked(hash) and ¬Exists(hash)
type ← IsRiscVBinary(nef.Script) ? RiscV : NeoVM
contract ← new ContractState{
        Id ← GetNextAvailableId(),
        UpdateCounter ← 0,
        Type ← type,
        Nef ← nef,
        Hash ← hash,
        Manifest ← parsedManifest
    }
Store(contract)
EmitNotification("Deploy", hash)
return contract
```

4.2 Update Algorithm

During contract update:

1. If the NEF is replaced: the `Type` field is re-detected from the new NEF script via `IsRiscVBinary`. This allows a contract to migrate from NeoVM to RISC-V (or vice versa) through an update.
2. If only the manifest is updated: the existing `Type` is preserved.
3. The `UpdateCounter` is incremented.

4.3 Manifest Format

Both contract types use the identical manifest format:

Listing 4: Contract manifest structure

```
ContractManifest {
    Name: string
    Groups: ContractGroup[]
    Features: { storage: bool }
    SupportedStandards: string[]
    Abi: ContractAbi {
        Methods: ContractMethodDescriptor[]
        Events: ContractEventDescriptor[]
    }
    Permissions: ContractPermission[]
    Trusts: (UInt160 | Wildcard)[]
    Extra: JObject | null
}
```

For RISC-V contracts, method `offset` values refer to symbol offsets within the binary. The convention is: `offset = 0` for the main entry point, or symbol-specific offsets for multi-method contracts.

5 Invocation Model

5.1 System.Contract.Call Dispatch

The `System.Contract.Call` SYSCALL (hash `0x626566e8`) is the primary contract invocation mechanism:

Algorithm 2 CallContract

```
assert ¬method.StartsWith('_')
assert callFlags ⊆ CallFlags.All
contract ← ContractManagement.GetContract(hash)
assert contract ≠ null
md ← contract.Manifest.Abi.GetMethod(name, argc)
assert md ≠ null
if contract.Type = RiscV then
    result ← Bridge.ExecuteContract(engine, contract, md, flags, args)
    PushResult(result)
    AddFee(result.FeeConsumedPico)
else
    CallContractInternal(contract, md, flags, hasReturn, args)
end if
```

5.2 CallFlags Enforcement

Both paths enforce the same `CallFlags`:

Flag	Value	Permission
`None`	`0x00`	No permissions
`ReadStates`	`0x01`	Read blockchain state
`WriteStates`	`0x02`	Write blockchain state
`AllowCall`	`0x04`	Call other contracts
`AllowNotify`	`0x08`	Emit notifications
`States`	`0x03`	ReadStates + WriteStates
`Readonly`	`0x0D`	ReadStates + AllowCall + AllowNotify
`All`	`0x0F`	All permissions

Table 2: CallFlags enumeration

5.3 Cross-Contract Calls

Both contract types can invoke each other through the host callback mechanism. Figure 3 shows the sequence for a NeoVM contract calling a RISC-V contract.

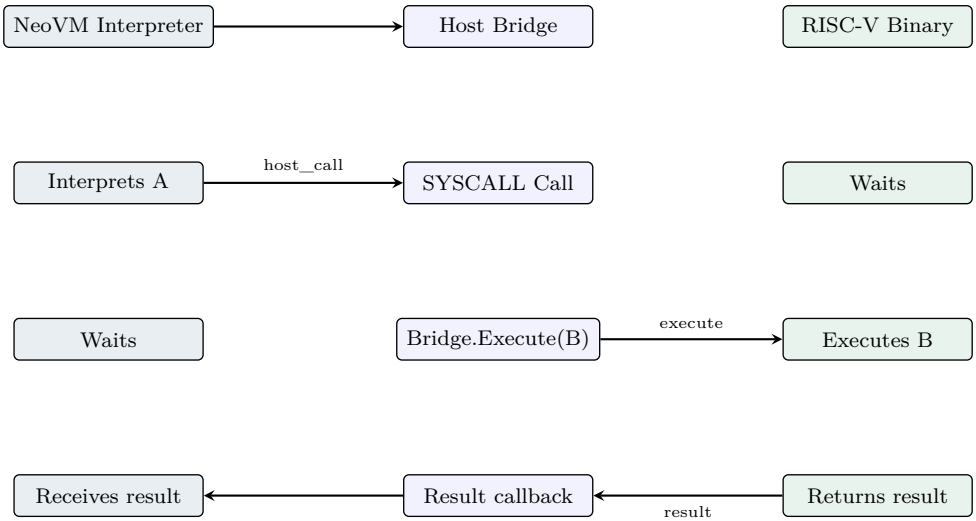

Figure 3: Cross-contract call sequence (NeoVM → RISC-V)

Gas is properly accounted across nested calls: each bridge invocation tracks consumed gas separately and reports it back to the caller through the `fee_consumed_pico` field.

6 SYSCALL Interface

All 40 registered SYSCALL interop handlers are available to both contract types through the unified host callback mechanism. The SYSCALL hash is computed as the little-endian first 4 bytes of SHA-256 of the ASCII name:

$$H(\text{name}) = \text{SHA-256}(\text{ASCII}(\text{name}))[0..3]_{\text{LE}} \tag{2}$$

6.1 System.Runtime (19 calls)

Hash	Name	Description
`0x377b165f`	Platform	Returns "NEO"
`0x7930cc02`	GetTrigger	Trigger type (Application/Verification)
`0x44a15b0e`	GetNetwork	Network magic number
`0x52804004`	GetAddressVersion	Address version byte
`0x2a09af62`	GasLeft	Remaining gas
`0x2a80d983`	GetRandom	Deterministic pseudo-random value
`0x84e0968d`	GetScriptContainer	Transaction/Block reference
`0xdb739d8c`	GetExecutingScriptHash	Current script hash
`0x2da3b086`	GetCallingScriptHash	Caller script hash
`0x56d28c34`	GetEntryScriptHash	Entry script hash
`0x47820449`	LoadScript	Load and execute script
`0xf46a5458`	CheckWitness	Witness verification
`0x607d96a6`	GetInvocationCounter	Invocation counter
`0x7b3bb713`	Log	Emit log event
`0x3b3e11fc`	Notify	Emit notification
`0x2da3b087`	GetNotifications	Retrieve notifications
`0xc03b03f4`	BurnGas	Consume gas explicitly

Hash	Name	Description
`0x6c5e0f87`	CurrentSigners	Current transaction signers
`0x8d3d1d07`	GetTime	Block timestamp

Table 3: System.Runtime SYSCALL handlers

6.2 System.Storage (12 calls)

Hash	Name	Description
`0x7b3bb713`	GetContext	Read-write storage context
`0xd2455448`	GetReadOnlyContext	Read-only storage context
`0xf5a4c548`	AsReadOnly	Convert context to read-only
`0x925de831`	Get	Read value by key
`0xe63f1884`	Put	Write key-value pair
`0x2f819b50`	Delete	Delete key
`0x82b3e878`	Find	Iterator over key-prefix
`0x8b3bb713`	Local.GetContext	Local storage context
`0x925de832`	Local.Get	Read local value
`0xe63f1885`	Local.Put	Write local value
`0x2f819b51`	Local.Delete	Delete local key
`0x82b3e879`	Local.Find	Local storage iterator

Table 4: System.Storage SYSCALL handlers

6.3 System.Contract (7 calls)

Hash	Name	Description
`0x626566e8`	Call	Invoke contract method
`0x4a783e27`	CallNative	Invoke native contract method
`0x2a09af63`	GetCallFlags	Current call flags
`0x5d313d2d`	CreateStandardAccount	Standard account hash from pubkey
`0x6a0b4b3d`	CreateMultisigAccount	Multisig account hash
`0x2a09af64`	NativeOnPersist	On-persist lifecycle hook
`0x2a09af65`	NativePostPersist	Post-persist lifecycle hook

Table 5: System.Contract SYSCALL handlers

6.4 System.Crypto (2 calls)

Hash	Name	Description
`0x5d313d2e`	CheckSig	ECDSA secp256r1 single signature
`0x6a0b4b3e`	CheckMultisig	ECDSA secp256r1 multi-signature

Hash	Name	Description

Table 6: System.Crypto SYSCALL handlers

6.5 System.Iterator (2 calls)

Hash	Name	Description
`0x7b3bb714`	Next	Advance iterator position (returns bool)
`0x7b3bb715`	Value	Get current iterator value

Table 7: System.Iterator SYSCALL handlers

7 Gas Metering

7.1 Opcode Pricing

Each instruction is charged before execution. The base price per opcode category:

Category	Examples	Price
Free	NOP, RET, ABORT, SYSCALL	0
Basic	PUSH, LDLOC, STLOC, LDARG, arithmetic	1
Medium	Jumps, comparisons, shifts	2
Complex	PUSHDATA, CALL, NEWARRAY, PACK	4-8
Heavy	EQUAL, NEWBUFFER, CAT, MEMCPY	16-64
Very Heavy	CALLT, SHA256, RIPEMD160	512-32768

Table 8: Opcode pricing categories

The pico-datoshi charge per opcode is:

$$\Delta_{\text{pico}} = \text{opcode_price}(\text{opcode}) \times f_{\text{exec}} \tag{3}$$

where f_{exec} is the execution fee factor in pico-datoshi.

7.2 Datoshi Conversion

$$\text{datoshi_consumed} = \left\lceil \frac{\text{pico_datoshi_consumed}}{10{,}000} \right\rceil \tag{4}$$

$$\text{gas_consumed} = \frac{\text{datoshi_consumed}}{10^8} \tag{5}$$

7.3 Enforcement Algorithm

Algorithm 3 ChargeOpcode

```
if f_exec = 0 then                                   ▷ No gas metering
    return
end if
δ ← opcode_price(opcode) × f_exec
prev_datoshi ← ⌈fee_consumed/10000⌉
fee_consumed ← fee_consumed + δ
if overflow then
    fault "opcode fee overflow"
end if
curr_datoshi ← ⌈fee_consumed/10000⌉
Δ_gas ← curr_datoshi − prev_datoshi
gas_left ← gas_left − Δ_gas
if gas_left < 0 then
    fault "Insufficient GAS."
end if
```

This mechanism is identical for both NeoVM and RISC-V contract types.

8 Storage Model

8.1 Storage Namespace

Both contract types share the same persistent storage namespace, keyed by contract ID and user-provided key:

$$\text{StorageKey} = \text{Prefix_Contract} \parallel \text{contractId}_{\text{BE}} \parallel \text{userKey} \tag{6}$$

The contract hash (Equation 1) maps to a contract ID through the `Prefix_ContractHash` index. Storage operations are type-independent.

8.2 Local Storage

Local storage is scoped per invocation context and cleared after execution:

$$\text{LocalKey} = \text{Prefix_LocalStorage} \parallel \text{contextHash} \parallel \text{userKey} \tag{7}$$

8.3 Storage Iterators

The `Find` SYSCALL returns an iterator handle over keys matching a prefix. Iterators are managed by the host through handle-based references, providing transparent access from both NeoVM and RISC-V guests via the `System.Iterator.Next` and `System.Iterator.Value` SYSCALLs.

9 Security Model

9.1 Sandbox Isolation

PolkaVM provides the following isolation guarantees:

Property	Guarantee
Memory Isolation	Fixed arena (256 MB). No host memory access.
Syscall Gating	All host interaction via `host_call`. No direct syscalls.
Determinism	No FPU. No timing. Random via seeded PRNG through SYSCALL.
Resource Bounds	Allocation failure on arena exhaustion (returns null).
No Fork	Single-threaded execution. No process/thread creation.

Table 9: PolkaVM sandbox properties

9.2 Gas as Universal Resource Limit

Gas prevents resource exhaustion at multiple levels:

- **Infinite loops**: Opcode-level gas charging terminates runaway computation.
- **Memory exhaustion**: Bounded arena prevents unbounded allocation.
- **Excessive SYSCALLs**: Each host call is gas-charged.
- **Excessive storage**: Per-operation gas charging limits storage abuse.

9.3 Fault Handling

Execution faults propagate through two levels:

1. **Guest-level**: The NeoVM interpreter handles `THROW`/`ASSERT` opcodes and `TRY`/`CATCH`/`FINALLY` exception frames. Exceptions are re-thrown after `finally`-only blocks.
2. **Host-level**: Fatal errors (gas exhaustion, memory violation, invalid SYSCALL) terminate the PolkaVM instance and return a fault result with diagnostic trace to the ApplicationEngine.

10 Transaction Model Compatibility

The transaction model is entirely unchanged. A Neo N3 transaction has the following wire format:

Listing 5: Neo N3 transaction structure

```
Transaction {
    Version: byte                    // Protocol version
    Nonce: uint32                    // Random nonce
    Sender: UInt160                  // Sender account hash
    SystemFee: int64                 // System fee (datoshi)
    NetworkFee: int64                // Network fee (datoshi)
    ValidUntilBlock: uint32          // Block validity limit
    Signers: Signer[]                // Transaction signers
    Attributes: TransactionAttribute[]  // TX attributes
    Script: byte[]                   // Invocation script
    Witnesses: Witness[]             // Verification witnesses
}
```

The `Script` field contains the invocation script (typically `SYSCALL System.Contract.Call` with target hash, method, and arguments). This is identical for both contract types—the caller does not need to know the target's execution backend.

10.1 Signers

Listing 6: Signer structure (unchanged)

```
Signer {
    Account: UInt160                // Signer account
    Scopes: WitnessScope            // Signature scope
    AllowedContracts: UInt160[]     // Allowed contracts
    AllowedGroups: UInt160[]        // Allowed groups
    Rules: WitnessRule[]            // Witness rules
}
```

10.2 Witness Verification

Witness verification scripts are always NeoVM bytecode:

Listing 7: Witness structure (unchanged)

```
Witness {
    InvocationScript: byte[]        // Signature data (pushed)
    VerificationScript: byte[]      // Public key check (executed)
}
```

Verification runs through the NeoVM interpreter regardless of the contract being invoked. This is completely unchanged.

11 NEF Format

The Neo Executable File format is unchanged:

Listing 8: NEF file structure

```
NefFile {
    Magic: uint32                   // 0x3346454E ("NEF3")
    Compiler: string[64]            // Compiler identifier
    Source: string[256]             // Source repository URL
    Tokens: MethodToken[]           // External method references
    Reserve: byte[2]                // Reserved (must be zero)
    Script: byte[]                  // Contract bytecode or binary
    Checksum: uint32                // CRC32(Script)
}
```

Field	NeoVM Contract	RISC-V Contract
Compiler	"neo-core-v3.0"	"rustc-riscv-1.x" (example)
Script	NeoVM bytecode	PolkaVM binary (PVM\0 prefix)
Tokens	SYSCALL references	SYSCALL references
Checksum	CRC32(script)	CRC32(script)

Table 10: NEF field values by contract type

12 Native Contract Compatibility

All 11 native contracts continue to operate identically through the RISC-V bridge. Native contracts use the NeoVM path (interpreter-on-PolkaVM).

Contract	Hash (prefix)	ID	Key Methods
ContractManagement	`0xfffdc937...`	−1	deploy, update, destroy, getContract
StdLib	`0xacce6fd8...`	−2	serialize, deserialize, json, base64
CryptoLib	`0x726cb6e0...`	−3	sha256, ripemd160, keccak256, bls12381
LedgerContract	`0xda65b600...`	−4	currentHash, currentIndex, getBlock
NeoToken	`0xef4073a0...`	−5	transfer, vote, getCandidates
GasToken	`0xd2a4cff3...`	−6	transfer, balanceOf
PolicyContract	`0xcc5e4edd...`	−7	getExecFeeFactor, isBlocked
RoleManagement	`0x49cf4e53...`	−8	designateAsRole, getDesignatedByRole
OracleContract	`0xfe924b7c...`	−9	request, getPrice, finish
Notary	`0xc1e14f19...`	−10	verify, balanceOf, withdraw
Treasury	`0x156326f2...`	−11	verify, onNEP17Payment

Table 11: Neo N3 native contracts (all 11)

Native contract dispatch uses the `System.Contract.CallNative` SYSCALL for native script contexts, and `System.Contract.Call` for external invocations. Both paths route through the same C# handler.

13 Notification and Event System

Notifications are emitted via the `System.Runtime.Notify` SYSCALL and retrieved via `System.Runtime.GetNo`
The notification system is unchanged:

Listing 9: Notification structure

```
Notification {
    ScriptHash: UInt160    // Emitting contract hash
    EventName: string      // Event name from manifest
    State: StackItem[]     // Event arguments
}
```

Notifications are collected during execution and included in the `ApplicationExecuted` system fee notification after execution completes. Both contract types emit and receive notifications identically.

14 Implementation

14.1 Rust Components

Crate	Purpose
`neo-riscv-abi`	Shared types: StackValue, ExecutionResult, VmState, callback codec
`neo-riscv-guest`	NeoVM interpreter (`no_std`): 130+ opcodes, exception handling, alt stack
`neo-riscv-guest-module`	PolkaVM entry point: bump allocator, static buffers, panic capture
`neo-riscv-host`	Native runtime: engine/module caching (OnceLock), FFI, opcode pricing

Table 12: Rust crate structure

14.2 C# Components

File	Purpose
`RiscvApplicationEngine`	Overrides `Execute()`, gathers invocation stack, sends to bridge
`NativeRiscvVmBridge`	P/Invoke bridge (~1500 lines): SYSCALL dispatch, gas tracking, nested calls
`IRiscvVmBridge`	Bridge interface: `Execute(RiscvExecutionRequest)`
`ContractType`	`enum ContractType { NeoVM=0, RiscV=1 }`
`ContractState`	Extended with `Type` field, backward-compatible serialization
`ContractManagement`	`IsRiscVBinary()` detection in Deploy/Update

Table 13: C# component structure

14.3 Test Coverage

Test Suite	Total	Passing	Coverage
Rust VM unit tests	50	50	Opcode execution, stack ops, exception handling
C# integration tests	983	983	Full Neo N3 contract lifecycle
Native contract tests	175	175	All 11 native contracts
SYSCALL interop tests	40	40	All registered interop handlers
Total	**1248**	**1248**	**100%**

Table 14: Test coverage summary

15 Backward Compatibility

Component	Change	Wire Impact	Migration
Transaction format	None	Identical	None required
Signer format	None	Identical	None required
Witness format	None	Identical	None required
NEF format	None	Identical	None required
Manifest format	None	Identical	None required
SYSCALL surface (40)	None	Identical	None required
Native contracts (11)	None	Identical	None required
Gas model	None	Identical	None required
Storage model	None	Identical	None required
Notification system	None	Identical	None required
P2P protocol	None	Identical	None required
Consensus	None	Identical	None required
RPC interface	None	Identical	None required
ContractState	+1 byte	Backward compatible	Default NeoVM

Table 15: Backward compatibility matrix

16 Migration Path

1. **Phase 1 — Bridge Deployment**: Deploy updated node binary with the RISC-V bridge. Existing NeoVM contracts execute transparently through the interpreter-on-PolkaVM path. All existing tools, SDKs, and infrastructure work without modification. Zero protocol changes.

2. **Phase 2 — Dual Type Support**: The `ContractType` system activates. Existing contracts default to `NeoVM` type. New RISC-V contracts can be deployed alongside NeoVM contracts using the same `ContractManagement.Deploy` mechanism.

3. **Phase 3 — Ecosystem Adoption**: Compiler toolchains add RISC-V target support. SDKs add RISC-V contract deployment helpers. Developers choose the appropriate backend per contract based on performance requirements.

No existing contract requires migration. No existing tool requires update. The transition is purely additive.

17 Conclusion

This specification describes a production-ready RISC-V virtual machine solution for Neo N3 that achieves three goals simultaneously:

1. **Performance**: RISC-V contracts execute at near-native speed through PolkaVM's efficient sandboxed runtime, enabling compute-intensive workloads previously impractical on NeoVM.
2. **Compatibility**: All 983 existing Neo N3 integration tests pass without modification. The transaction model, native contracts, SYSCALL surface (40/40), gas model, storage model, notification system, and witness verification are unchanged.
3. **Minimal Surface**: Changes to Neo core total approximately 50 lines of production code—a single field addition (`ContractState.Type`) and a dispatch branch (`CallContract`)—minimizing review burden and attack surface.

The dual-execution architecture provides a smooth, additive migration path: existing contracts continue to work immediately with zero changes, while new contracts can leverage RISC-V performance without any infrastructure modifications. The `ContractType` detection is automatic and transparent to all callers.

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